Kirill Kozlov

Junior Python Backend Developer (FastAPI, PostgreSQL, Kubernetes)

Moscow, Russia | Phone: +7 (910) 968 1153 | Email: kirillkozlov.yar@gmail.com
GitHub: github.com/noobmaster-ru | Telegram: @noobmaster_rus

Professional Summary

Software engineer and Machine Learning master's student at CMC MSU with production experience at X5 Tech. Built Python services using FastAPI, PostgreSQL, and Kubernetes; developing AI product search and catalog-processing pipelines for Korzina.

Technical Skills

Programming Languages: Python, SQL, Go, C++
Frameworks: FastAPI, SQLAlchemy (async), Alembic, Pydantic, aiogram, pytest
Databases and Messaging: PostgreSQL, Redis, Apache Kafka
Infrastructure and DevOps: Docker, Kubernetes (K8s), Helm, GitLab CI/CD, Nginx, Git, Linux
Libraries: asyncio, httpx, aiohttp, requests, pandas, NumPy, scikit-learn, structlog
Artificial Intelligence and Machine Learning: Large Language Models (LLM), LLM agents, RAG (Retrieval-Augmented Generation), BM25, OpenAI API, LangChain, prompt engineering, OCR (Optical Character Recognition), A/B testing, PyTorch, Claude Code
Languages: Russian — Native, English — Upper-Intermediate (B2)

Work Experience

Python Backend Developer (Junior) | X5 Tech (X5 Group, Russia's largest grocery retailer), Logistics Solutions Department, Moscow | October 2026 — Present

- Developing the system that handles 90% of all foreign supplies X5 Retail Group purchases for its retail chains.

Python Backend Developer (Intern) | X5 Tech, AI Agents Department, Moscow | April 2026 — October 2026

- Developed a FastAPI and PostgreSQL (SQLAlchemy async, Alembic) backend service for an LLM contract-review agent: over 500 production reviews in May–August 2026; business impact of 8 FTE for X5 Group.
- Cut legal opinion preparation time to 8–19 minutes versus about 97 minutes of manual lawyer work.
- Developed the FastAPI backend for a utility-bill processing service: consuming OCR results from Apache Kafka, storing them and serving them to the frontend; business impact of 10 FTE.
- Built and deployed to dev and prod a stateless REST API for foreign invoice analysis (OCR and LLM) for X5 Tech; in pilot operation since August 2026.
- Deployed the services to a Kubernetes cluster (dev and prod): Helm charts, image build and deploy in GitLab CI/CD, latency metrics in Grafana.

Projects

Korzina — Marketplace and AI Product Search | TypeScript, Node.js, PostgreSQL, React | September 2026 — Present

- Building a Telegram Mini App and bot with a catalog of 100,000+ products; automated supplier-data ingestion, normalization, and deduplication.
- Implemented PostgreSQL full-text and trigram search; developing a hybrid retrieval prototype combining multimodal Gemini embeddings, pgvector, reciprocal rank fusion (RRF), and Qwen reranking.
- Implemented LLM/VLM attribute extraction and product-compatibility checks in the prototype, with versioned embedding caches, concurrent-request deduplication, and incremental refresh.
- Built offline relevance evaluation with frozen datasets, qrels, Recall@20, and Precision@5; compared text and multimodal retrieval under model-cost and latency constraints.
- Built the Fastify API and background jobs; deployed the service using PostgreSQL, Valkey, S3, Docker, and GitLab CI/CD.

Other projects: AxiomAI: LLM-powered Telegram bot, 80% reduction in manager workload; WB Parser: Wildberries parser and Mini App, 220 users; Placa: analytics for 300 sellers.

Education

Lomonosov Moscow State University, Master's degree | Faculty of Computational Mathematics and Cybernetics, Department of Mathematical Methods of Forecasting, major in Machine Learning, Moscow | September 2026 — Present

Lomonosov Moscow State University, Bachelor's degree | Faculty of Computational Mathematics and Cybernetics, Moscow | August 2022 — August 2026

- Deep Learning School (DLS), Computer Vision and NLP tracks — MIPT School of Deep Learning.
- Summer School on A/B Testing, X5 Tech — Summer 2026 (certificate no. 9074503).